

Lab 4 Tuning ASR and TTS

Part A: Hard cases for Speech Recognition

I added the below lines to print utterance and confidence.

```
console.log("Utterance:", event.value[0].utterance)
console.log("Confidence:", event.value[0].confidence)
```

I tried checking the confidence score of various words. The results of some are below

- Popular fictional names like Hermione Granger, Daenerys Targaryen scored high confidence score.
- Some words improved with repeated attempts like Demogorgon whereas Indian, Arabic and French names scored consistently low even after multiple attempts.
- Interestingly, "Hagride" a word from old English was captured as "Hagrid" a Harry Potter character.
- Azure automatically censors certain words and replaces them with ****, as seen with Homo sapiens.

Category	Word	ASR Heard	Confidence	Confidence after custom speech training
Fictional	Hermione Granger	Hermione Granger	0.7218849	0.8604097
Fictional	Daenerys Targaryen	Daenerys Targaryen	0.68041193	0.75845146
English name	John Jacob	John Jacob	0.62918675	
Indian city	Thiruvananthapuram	Thiruvananthapuram	0.4658637	0.38524365
Old English	Hagride (old English)	Hagrid	0.3885026	0.7258304
Arabic name	Sheikh Zayed bin Sultan Al Nahyan	Sheikh said Sultan Al Nahyan	0.3439688	0.5264717
Scientific	Homo sapiens	**** sapien	0.36796866	0.84268826
Fictional	Demogorgon	Demogorgon	0.21270892	0.2675049
Botanical	Bougainvillea	Bougainvillea	0.15447584	0.6748224
Indian city	Trichambaram	Trichambaram	0.15417889	0.19269605
Indian name	Mamitha Muraleedharan	Mamita Muralidharan	0.11488922	0.1793807
Fictional	Dementors	Dementos	0.12164514	0.3368625
Arabic city	Khorfakkan	Karfakan	0.06516775	0.10117695
French city	Megève	Midgev	0.045654885	0.065107584

Based on my experiments, confidence scores were consistently higher for Western English names like John Jacob compared to Indian, Arabic and French names. As a non native English speaker, my own name scored as low as 0.11, suggesting that accents significantly affects recognition. French and Arabic city names scored the worst (Megève: 0.05, Khorfakkan: 0.07). A possible solution is Azure Custom Speech, where custom training data with specific words and accents can improve recognition accuracy.

Part A-VG: Azure Custom Speech

Endpoint ID = 459f4d2a-49c1-457e-84fd-465b7dc62042

The custom model showed notable improvements for several words, particularly Homo sapiens (0.37 to 0.84) and Bougainvillea (0.15 to 0.67). Homo sapiens was no longer censored by Azure after custom

training. However, French and Arabic place names like Megève and Khorfakkan showed minimal improvement. This is likely because I used text file for training rather than audio files. Audio files might have helped with the pronunciation of words which are said with certain accent. Hence it would have helped to get better results for words like Megève and Khorfakkan.